Matrix Analysis Prerequisites

Prerequisites for the formalisation of Vern Paulsen's lectures on quantum computing

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Abstract

General facts about complex matrices that the formalised lectures (session `Quantum_Computing_CP_Maps`) build on: sums, adjoints, unitary matrices and Kronecker products, linear-space isomorphisms, and matrices of fixed dimension as a type, with the Hilbert–Schmidt norm (a normed space for the Hahn–Banach theorem of `HOL-Hahn_Banach`) and the structure of a Euclidean space (so that `HOL-Analysis` applies). Nothing here depends on the lectures.

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The gate names of Isabelle_Marries_Dirac would capture the letters X, Y, Z, H, S, T used as variables in the lectures.

hide-const (**open**) *Quantum.X Quantum.Y Quantum.Z Quantum.H Quantum.S Quantum.T*

1 Complex numbers and scalar multiples

lemma *cnj-mult-self-cmod*: *cnj z * z = complex-of-real ((cmod z)²)*
⟨*proof*⟩

lemma *smult-smult-mat*:
fixes *A* :: *complex Matrix.mat*
shows *a ·m (b ·m A) = (a * b) ·m A*
⟨*proof*⟩

lemma *zero-smult-mat-carrier*:
fixes *A* :: *complex Matrix.mat*
assumes *A ∈ carrier-mat m k*
shows *0 ·m A = 0m m k*
⟨*proof*⟩

2 The adjoint

The adjoint of an $m \times n$ matrix is $n \times m$.

lemma *dagger-carrier*:
assumes *M ∈ carrier-mat m n*
shows *(M†) ∈ carrier-mat n m*
⟨*proof*⟩

lemma *dagger-mult-inner*:
fixes *Y* :: *complex Matrix.mat*
assumes *Yc*: *Y ∈ carrier-mat p q* **and** *i*: *i < q* **and** *j*: *j < q*
shows *((Y†) * Y) $$ (i,j) = ⟨Matrix.col Y i | Matrix.col Y j⟩*
⟨*proof*⟩

3 Sums of vectors

Finite sums of vectors, entrywise.

definition *vec-sum* :: *nat ⇒ 'c set ⇒ ('c ⇒ complex Matrix.vec) ⇒ complex Matrix.vec* **where**
vec-sum n Js w = Matrix.vec n (λi. sum (λJ. w J $ i) Js)

lemma *vec-sum-dim* [*simp*]: *dim-vec (vec-sum n Js w) = n*
⟨*proof*⟩

lemma *vec-sum-index* [*simp*]: *i < n ⟹ vec-sum n Js w $ i = sum (λJ. w J $ i) Js*
⟨*proof*⟩

lemma *vec-sum-cong*:
assumes *⋀J. J ∈ Js ⟹ w J = w' J*
shows *vec-sum n Js w = vec-sum n Js w'*
⟨*proof*⟩

lemma *mult-mat-vec-vec-sum*:
assumes *Ac*: *A ∈ carrier-mat m n* **and** *fin*: *finite Js*
and *wd*: *⋀J. J ∈ Js ⟹ dim-vec (w J) = n*

shows $A *_v$ *vec-sum n Js w* = *vec-sum m Js* (λJ. $A *_v w J$)
⟨*proof*⟩

lemma *mult-mat-vec-smult-vec*:
fixes A :: *complex Matrix.mat* **and** v :: *complex Matrix.vec*
assumes *Ac*: $A \in$ *carrier-mat m n* **and** *vd*: *dim-vec v* = *n*
shows $A *_v (c \cdot_v v) = c \cdot_v (A *_v v)$
⟨*proof*⟩

4 Products and sums of matrices

lemma *mat-mult-entry*:
fixes A B :: *complex Matrix.mat*
assumes $A \in$ *carrier-mat r s* $B \in$ *carrier-mat s t* $k < r$ $j < t$
shows ($A * B$) $$ (k,j) = ($\sum l$<*s*. A $$ (k,l) * B $$ (l,j))
⟨*proof*⟩

The same with the sum over $\{0.. < q\}$; square matrices that agree on every vector are equal; a non-zero matrix has a non-zero entry.

lemma *mat-mult-index-sum*:
assumes *A-c*: $A \in$ *carrier-mat p q*
and *B-c*: $B \in$ *carrier-mat q s*
and *r-lt*: $r < p$ **and** *c-lt*: $c < s$
shows ($A * B$) $$ (r, c) = ($\sum k$ = *0*..<*q*. A $$ (r, k) * B $$ (k, c))
⟨*proof*⟩

lemma *mat-eq-of-mult-vec-eq*:
fixes M N :: *complex Matrix.mat*
assumes *M-c*: $M \in$ *carrier-mat k k*
and *N-c*: $N \in$ *carrier-mat k k*
and *eq*: $\bigwedge x$. $x \in$ *carrier-vec k* $\Longrightarrow$ $M *_v x = N *_v x$
shows $M = N$
⟨*proof*⟩

lemma *nonzero-entry*:
fixes A :: *complex Matrix.mat*
assumes A: $A \in$ *carrier-mat m k* **and** *nz*: $A \neq 0_m$ *m k*
obtains *r s* **where** $r < m$ $s < k$ A $$ (r,s) $\neq$ *0*
⟨*proof*⟩

An independent frame cancels: if M^*M is invertible, $MAM^* = 0$ forces $A = 0$.

lemma *frame-cancel*:
fixes M A B :: *complex Matrix.mat*
assumes M: $M \in$ *carrier-mat m k* **and** A: $A \in$ *carrier-mat k k* **and** B: $B \in$ *carrier-mat k k*
and *BG*: $B * (M^\dagger * M) = 1_m$ *k* **and** *GB*: $(M^\dagger * M) * B = 1_m$ *k*
and *z*: $M * A * (M^\dagger) = 0_m$ *m m*
shows $A = 0_m$ *k k*
⟨*proof*⟩

lemma *foldr-add-mat-carrier*:
fixes f :: $'a \Rightarrow$ *complex Matrix.mat*
assumes $\forall x \in$ *set xs*. $f x \in$ *carrier-mat n n*
shows *foldr* (λx *acc*. $f x$ + *acc*) *xs* (0_m *n n*) $\in$ *carrier-mat n n*
⟨*proof*⟩

lemma *foldr-add-mat-index*:
fixes f :: $'a \Rightarrow$ *complex Matrix.mat*

assumes $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ n\ n$ **and** $i < n$ **and** $j < n$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ n\ n)\ \$\$\ (i,\ j) = sum\text{-}list\ (map\ (\lambda x.\ f\ x\ \$\$\ (i,\ j))\ xs)$
⟨*proof*⟩

lemma *foldr-add-mat-carrier-rect*:
assumes $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ p\ q$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ p\ q) \in carrier\text{-}mat\ p\ q$
⟨*proof*⟩

lemma *foldr-add-mat-index-rect*:
assumes $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ p\ q$ **and** $i < p$ **and** $j < q$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ p\ q)\ \$\$\ (i,j) = (\sum x \leftarrow xs.\ f\ x\ \$\$\ (i,j))$
⟨*proof*⟩

Carrier-preservation in a different dimension pair.

lemma *foldr-add-carrier-dn*:
fixes $f :: 'a \Rightarrow complex\ Matrix.mat$
assumes $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ d\ n$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ d\ n) \in carrier\text{-}mat\ d\ n$
⟨*proof*⟩

lemma *foldr-add-mat-cong*:
fixes $f\ g :: 'a \Rightarrow complex\ Matrix.mat$
assumes $\bigwedge x.\ x \in set\ xs \Longrightarrow f\ x = g\ x$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ z = foldr\ (\lambda x\ acc.\ g\ x + acc)\ xs\ z$
⟨*proof*⟩

Sums over $[0..<N]$, and their entries.

lemma *foldr-upt-add-acc-index*:
fixes $f :: nat \Rightarrow complex\ Matrix.mat$
assumes *carr*: $\forall k < N.\ f\ k \in carrier\text{-}mat\ d\ d$
and *z-c*: $z \in carrier\text{-}mat\ d\ d$
and *r-lt*: $r < d$ **and** *c-lt*: $c < d$
shows $foldr\ (\lambda k\ acc.\ f\ k + acc)\ [0..<N]\ z\ \$\$\ (r,\ c)$
$= (\sum k = 0..<N.\ f\ k\ \$\$\ (r,\ c)) + z\ \$\$\ (r,\ c)$
⟨*proof*⟩

lemma *foldr-upt-add-index*:
fixes $f :: nat \Rightarrow complex\ Matrix.mat$
assumes *carr*: $\forall k < N.\ f\ k \in carrier\text{-}mat\ d\ d$
and *r-lt*: $r < d$ **and** *c-lt*: $c < d$
shows $foldr\ (\lambda k\ acc.\ f\ k + acc)\ [0..<N]\ (0_m\ d\ d)\ \$\$\ (r,\ c)$
$= (\sum k = 0..<N.\ f\ k\ \$\$\ (r,\ c))$
⟨*proof*⟩

lemma *foldr-smult-index*:
assumes *Bs*: $\forall B \in set\ Bs.\ B \in carrier\text{-}mat\ n\ m$ **and** *r*: $r < n$ **and** *s*: $s < m$
shows $foldr\ (\lambda k\ acc.\ c\ k \cdot_m (Bs\ !\ k) + acc)\ [0..<length\ Bs]\ (0_m\ n\ m)\ \$\$\ (r,s)$
$= (\sum k<length\ Bs.\ c\ k * Bs\ !\ k\ \$\$\ (r,s))$
⟨*proof*⟩

4.1 Algebra of *foldr*-sums of matrices

lemma *foldr-add-mat-mult-left*:
fixes $f :: 'a \Rightarrow complex\ Matrix.mat$
assumes *Ac*: $A \in carrier\text{-}mat\ m\ n$ **and** *fc*: $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ n\ p$
shows $A * foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ n\ p) = foldr\ (\lambda x\ acc.\ A * f\ x + acc)\ xs\ (0_m\ m\ p)$
⟨*proof*⟩

lemma *foldr-add-mat-mult-right*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat*
assumes *Bc*: $B \in$ *carrier-mat n p* **and** *fc*: $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat m n*
shows *foldr* $(\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ m\ n) * B =$ *foldr* $(\lambda x\ acc.\ f\ x * B + acc)\ xs\ (0_m\ m\ p)$
⟨*proof*⟩

lemma *foldr-add-mat-dagger*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat*
assumes $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat m n*
shows $(foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ m\ n))^\dagger = foldr\ (\lambda x\ acc.\ (f\ x)^\dagger + acc)\ xs\ (0_m\ n\ m)$
⟨*proof*⟩

lemma *foldr-add-mat-zero*:
foldr $(\lambda x\ acc.\ (0_m\ m\ n ::$ *complex Matrix.mat*$) + acc)\ xs\ (0_m\ m\ n) = 0_m\ m\ n$
⟨*proof*⟩

lemma *foldr-add-mat-delta*:
fixes $f ::$ *nat* $\Rightarrow$ *complex Matrix.mat*
assumes *fc*: $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat m n* **and** *d*: *distinct xs* **and** *i*: $i \in$ *set xs*
shows *foldr* $(\lambda x\ acc.\ (if\ x = i\ then\ f\ x\ else\ 0_m\ m\ n) + acc)\ xs\ (0_m\ m\ n) = f\ i$
⟨*proof*⟩

lemma *foldr-add-mat-smult-pull*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat* **and** $c :: 'a \Rightarrow$ *complex*
assumes *fc*: $\forall x \in$ *set xs*. $f\ x = c\ x \cdot_m M$ **and** *Mc*: $M \in$ *carrier-mat m n*
shows *foldr* $(\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ m\ n) =$ *sum-list* $(map\ c\ xs) \cdot_m M$
⟨*proof*⟩

lemma *foldr-add-mat-smult*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat*
assumes $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat m n*
shows *foldr* $(\lambda x\ acc.\ a \cdot_m f\ x + acc)\ xs\ (0_m\ m\ n) = a \cdot_m foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ m\ n)$
⟨*proof*⟩

lemma *foldr-list-as-index*:
fixes $g ::$ *complex Matrix.mat* $\Rightarrow$ *complex Matrix.mat*
shows *foldr* $(\lambda A\ acc.\ g\ A + acc)\ As\ z =$ *foldr* $(\lambda l\ acc.\ g\ (As\ !\ l) + acc)\ [0..<length\ As]\ z$
⟨*proof*⟩

definition *msum* $::$ *nat* $\Rightarrow$ *nat* $\Rightarrow$ $('a \Rightarrow$ *complex Matrix.mat*$) \Rightarrow 'a$ *list* $\Rightarrow$ *complex Matrix.mat* **where**
msum r c g xs $=$ *foldr* $(\lambda x\ acc.\ g\ x + acc)\ xs\ (0_m\ r\ c)$

lemma *msum-Nil* [*simp*]: *msum r c g* $[] = 0_m\ r\ c$
⟨*proof*⟩

lemma *msum-Cons* [*simp*]: *msum r c g* $(x\ \#\ xs) = g\ x +$ *msum r c g xs*
⟨*proof*⟩

lemma *msum-carrier*: $\forall x \in$ *set xs*. $g\ x \in$ *carrier-mat r c* $\Longrightarrow$ *msum r c g xs* $\in$ *carrier-mat r c*
⟨*proof*⟩

lemma *msum-index*:
assumes $\forall x \in$ *set xs*. $g\ x \in$ *carrier-mat r c* **and** $i < r$ **and** $j < c$
shows *msum r c g xs* \$\$ $(i,j) = (\sum x \leftarrow xs.\ g\ x$ \$\$ $(i,j))$
⟨*proof*⟩

Every *msum* has its stated dimensions; multiplying a sum on the right.

lemma *msum-dims* [*simp*]: *dim-row* (*msum r c g xs*) = *r dim-col* (*msum r c g xs*) = *c*
⟨*proof*⟩

lemma *msum-mult-right*:
assumes ∀ *x* ∈ *set xs*. *g x* ∈ *carrier-mat r c* **and** *B* ∈ *carrier-mat c p*
shows *msum r c g xs* ∗ *B* = *msum r p* (λ*x*. *g x* ∗ *B*) *xs*
⟨*proof*⟩

lemma *msum-mult-left*:
assumes *A* ∈ *carrier-mat m r* **and** ∀ *x* ∈ *set xs*. *g x* ∈ *carrier-mat r c*
shows *A* ∗ *msum r c g xs* = *msum m c* (λ*x*. *A* ∗ *g x*) *xs*
⟨*proof*⟩

Sums of matrices over a list: congruence, additivity, scalars, concatenation.

lemma *msum-cong*: (⋀*x*. *x* ∈ *set xs* ⟹ *g x* = *h x*) ⟹ *msum r c g xs* = *msum r c h xs*
⟨*proof*⟩

lemma *msum-add*:
assumes *g*: ∀ *x* ∈ *set xs*. *g x* ∈ *carrier-mat r c* **and** *h*: ∀ *x* ∈ *set xs*. *h x* ∈ *carrier-mat r c*
shows *msum r c* (λ*x*. *g x* + *h x*) *xs* = *msum r c g xs* + *msum r c h xs*
⟨*proof*⟩

lemma *msum-smult*:
assumes *g*: ∀ *x* ∈ *set xs*. *g x* ∈ *carrier-mat r c*
shows *msum r c* (λ*x*. *a* ·$_m$ *g x*) *xs* = *a* ·$_m$ *msum r c g xs*
⟨*proof*⟩

lemma *msum-append*:
assumes *g*: ∀ *x* ∈ *set xs*. *g x* ∈ *carrier-mat r c* **and** *g'*: ∀ *x* ∈ *set ys*. *g x* ∈ *carrier-mat r c*
shows *msum r c g* (*xs* @ *ys*) = *msum r c g xs* + *msum r c g ys*
⟨*proof*⟩

5 Finite sums and index arithmetic

lemma *sum-list-product-upt*:
(∑ *x*←*List.product* [*0*..<*a*] [*0*..<*b*]. *h x*) = (∑ *i*<*a*. ∑ *j*<*b*. *h* (*i*,*j*))
⟨*proof*⟩

lemma *sum-delta2*:
fixes *a d b e* :: *nat*
assumes *a*: *a* < *d* **and** *b*: *b* < *e*
shows (∑ *i*<*d*. ∑ *j*<*e*. *if i* = *a* ∧ *j* = *b then v else 0*) = *v*
⟨*proof*⟩

A list of 0s and 1s summing to 1 has exactly one 1.

lemma *sum-zero-one-eq-one*:
fixes *D* :: *real list*
assumes *D01*: ⋀*i*. *i* < *n* ⟹ *D* ! *i* = *0* ∨ *D* ! *i* = *1* **and** *D*: *length D* = *n* **and** *s*: *sum-list D* = *1*
shows ∃ *j*<*n*. *D* ! *j* = *1* ∧ (∀ *i*<*n*. *i* ≠ *j* ⟶ *D* ! *i* = *0*)
⟨*proof*⟩

lemma *Re-of-real-eq*: *complex-of-real r* = *z* ⟹ *r* = *Re z*
⟨*proof*⟩

A number is determined by its quotient and remainder.

lemma *nat-eq-iff-div-mod*:
fixes *x y k* :: *nat*

shows $x = y \longleftrightarrow x \; div \; k = y \; div \; k \wedge x \; mod \; k = y \; mod \; k$
⟨*proof*⟩

Concatenating rows of equal length, and the reindexing $\kappa = ik + j$.

lemma *concat-uniform-length*:
assumes $\forall xs \in set \; L. \; length \; xs = m$
shows $length \; (concat \; L) = length \; L * m$
⟨*proof*⟩

lemma *concat-uniform-nth*:
assumes *uniform*: $\forall xs \in set \; L. \; length \; xs = m$
and *i-lt*: $i < length \; L$
and *j-lt*: $j < m$
shows $concat \; L \; ! \; (i * m + j) = L \; ! \; i \; ! \; j$
⟨*proof*⟩

lemma *sum-split-prod-atLeastLessThan*:
fixes $f :: nat \Rightarrow 'a::comm\text{-}monoid\text{-}add$
shows $(\sum \kappa = 0..<N * k. \; f \; \kappa) = (\sum i = 0..<N. \; \sum j = 0..<k. \; f \; (i * k + j))$
⟨*proof*⟩

Sums over pairs of indices (i, j) with $i, j < k$.

lemma *msum-pairs-carrier*:
assumes $\bigwedge i \; j. \; i < k \Longrightarrow j < k \Longrightarrow g \; i \; j \in carrier\text{-}mat \; r \; c$
shows $msum \; r \; c \; (\lambda(i, \; j). \; g \; i \; j) \; (List.product \; [0..<k] \; [0..<k]) \in carrier\text{-}mat \; r \; c$
⟨*proof*⟩

lemma *msum-pairs-index*:
assumes *g*: $\bigwedge i \; j. \; i < k \Longrightarrow j < k \Longrightarrow g \; i \; j \in carrier\text{-}mat \; r \; c$ **and** *a*: $a < r$ **and** *b*: $b < c$
shows $msum \; r \; c \; (\lambda(i, \; j). \; g \; i \; j) \; (List.product \; [0..<k] \; [0..<k]) \; \$\$ \; (a, \; b)$
$= (\sum i{<}k. \; \sum j{<}k. \; g \; i \; j \; \$\$ \; (a, \; b))$
⟨*proof*⟩

lemma *msum-pairs-mult-right*:
assumes *g*: $\bigwedge i \; j. \; i < k \Longrightarrow j < k \Longrightarrow g \; i \; j \in carrier\text{-}mat \; r \; c$ **and** *B*: $B \in carrier\text{-}mat \; c \; p$
shows $msum \; r \; c \; (\lambda(i, \; j). \; g \; i \; j) \; (List.product \; [0..<k] \; [0..<k]) * B$
$= msum \; r \; p \; (\lambda(i, \; j). \; g \; i \; j * B) \; (List.product \; [0..<k] \; [0..<k])$
⟨*proof*⟩

lemma *msum-pairs-mult-left*:
assumes *g*: $\bigwedge i \; j. \; i < k \Longrightarrow j < k \Longrightarrow g \; i \; j \in carrier\text{-}mat \; r \; c$ **and** *A*: $A \in carrier\text{-}mat \; m \; r$
shows $A * msum \; r \; c \; (\lambda(i, \; j). \; g \; i \; j) \; (List.product \; [0..<k] \; [0..<k])$
$= msum \; m \; c \; (\lambda(i, \; j). \; A * g \; i \; j) \; (List.product \; [0..<k] \; [0..<k])$
⟨*proof*⟩

lemma *msum-upt-index*:
assumes $\bigwedge x. \; x < m \Longrightarrow g \; x \in carrier\text{-}mat \; r \; c$ **and** $a < r$ **and** $b < c$
shows $msum \; r \; c \; g \; [0..<m] \; \$\$ \; (a, \; b) = (\sum x{<}m. \; g \; x \; \$\$ \; (a, \; b))$
⟨*proof*⟩

lemma *idx8*: $k < (8::nat) \Longrightarrow k = 0 \vee k = 1 \vee k = 2 \vee k = 3 \vee k = 4 \vee k = 5 \vee k = 6 \vee k = 7$
⟨*proof*⟩

6 Linear maps on a set of matrices

Linearity of a map defined on a set of $n \times n$ matrices (an operator system, from Day 24 on) with values in M_d.

definition *lin-on* :: *nat* ⇒ *nat* ⇒ *complex Matrix.mat set* ⇒ (*complex Matrix.mat* ⇒ *complex Matrix.mat*) ⇒ *bool* **where**
lin-on n d OS Φ ⟷ (∀ *A* ∈ *OS*. Φ *A* ∈ *carrier-mat d d*) ∧ (∀ *A* ∈ *OS*. ∀ *B* ∈ *OS*. Φ (*A* + *B*) = Φ *A* + Φ *B*)
∧ (∀ *A* ∈ *OS*. ∀ *c*. Φ (*c* ·$_m$ *A*) = *c* ·$_m$ Φ *A*)

On a set closed under sums and containing 0, such a map commutes with finite sums.

lemma *msum-closed*:
assumes *S0*: 0_m *n n* ∈ *S* **and** *S*: ⋀*A B*. *A* ∈ *S* ⟹ *B* ∈ *S* ⟹ *A* + *B* ∈ *S*
and *g*: ∀ *x* ∈ *set xs*. *g x* ∈ *S*
shows *msum n n g xs* ∈ *S*
⟨*proof*⟩

lemma *lin-on-msum-closed*:
assumes *L*: *lin-on n d S* Φ **and** *S0*: 0_m *n n* ∈ *S* **and** *S*: ⋀*A B*. *A* ∈ *S* ⟹ *B* ∈ *S* ⟹ *A* + *B* ∈ *S*
and *g*: ∀ *x* ∈ *set xs*. *g x* ∈ *S*
shows Φ (*msum n n g xs*) = *msum d d* (λ*x*. Φ (*g x*)) *xs*
⟨*proof*⟩

lemma *lin-on-smult*:
assumes *lin-on n d S F*
shows *lin-on n d S* (λ*X*. *a* ·$_m$ *F X*)
⟨*proof*⟩

7 Linear-space isomorphisms

Matrix spaces are represented by carrier predicates on the single type *complex mat*. The record below packages a carrier with its zero, addition and scalar multiplication, so that statements such as $L(\mathcal{H}_A) \otimes L(\mathcal{H}_B) \cong L(\mathcal{H}_A \otimes \mathcal{H}_B)$ (Day 15) can be made as explicit linear isomorphisms with explicit inverses.

record ('*a*, '*v*) *linear-structure* =
ls-carrier :: '*v set*
ls-zero :: '*v*
ls-add :: '*v* ⇒ '*v* ⇒ '*v*
ls-smul :: '*a* ⇒ '*v* ⇒ '*v*

definition *linear-structure-map* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ ('*v* ⇒ '*w*) ⇒ *bool* **where**
linear-structure-map V W f ≡
f (*ls-zero V*) = *ls-zero W* ∧
(∀ *x* ∈ *ls-carrier V*. *f x* ∈ *ls-carrier W*) ∧
(∀ *x* ∈ *ls-carrier V*. ∀ *y* ∈ *ls-carrier V*.
f (*ls-add V x y*) = *ls-add W* (*f x*) (*f y*)) ∧
(∀ *c x*. *x* ∈ *ls-carrier V* ⟶
f (*ls-smul V c x*) = *ls-smul W c* (*f x*))

definition *is-isomorphism-between* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ ('*v* ⇒ '*w*) ⇒ *bool* **where**
is-isomorphism-between V W f ≡
linear-structure-map V W f ∧
(∃ *g*. *linear-structure-map W V g* ∧
(∀ *x* ∈ *ls-carrier V*. *g* (*f x*) = *x*) ∧
(∀ *y* ∈ *ls-carrier W*. *f* (*g y*) = *y*))

definition *is-isomorphic-to* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ *bool* **where**

is-isomorphic-to V W ≡ ∃ *f*. *is-isomorphism-between V W f*

lemma *is-isomorphism-betweenD-map*:
assumes *is-isomorphism-between V W f*
shows *linear-structure-map V W f*
⟨*proof*⟩

lemma *is-isomorphism-betweenD-inverse*:
assumes *is-isomorphism-between V W f*
obtains *g* **where** *linear-structure-map W V g*
and ⋀*x*. *x* ∈ *ls-carrier V* ⟹ *g* (*f x*) = *x*
and ⋀*y*. *y* ∈ *ls-carrier W* ⟹ *f* (*g y*) = *y*
⟨*proof*⟩

The example used throughout: $M_{m,n}$ with entrywise addition and scaling.

definition *mat-linear-structure* :: *nat* ⇒ *nat* ⇒ (*complex*, *complex Matrix.mat*) *linear-structure* **where**
mat-linear-structure m n ≡
(‖ *ls-carrier* = *carrier-mat m n*,
ls-zero = 0_m *m n*,
ls-add = (λ*x y*. *x* + *y*),
ls-smul = (λ*c x*. *c* ·$_m$ *x*) ‖)

hide-const (**open**) *Quantum.X Quantum.Y Quantum.Z Quantum.H Quantum.S Quantum.T*
declare *cauchy-schwarz-eq* [*simp del*] *cauchy-schwarz-col* [*simp del*] *eq-from-inner-prod* [*simp del*]

8 Unitary algebra

Library-level consequences of unitarity ($U^\dagger U = I$ and $UU^\dagger = I$ in the library's definition) used throughout; Paulsen's Day 3 characterisations of unitaries are in theory Day03_Unitary_Matrices.

lemma *unitary-square-dim*:
assumes *uA*: *unitary A*
shows *dim-row A* = *dim-col A*
⟨*proof*⟩

lemma *unitary-inv-l-dimcol*:
assumes *uA*: *unitary A*
shows ($A^\dagger$) ∗ *A* = 1_m (*dim-col A*)
⟨*proof*⟩

lemma *unitary-inv-r-dimrow*:
assumes *uA*: *unitary A*
shows *A* ∗ ($A^\dagger$) = 1_m (*dim-row A*)
⟨*proof*⟩

lemma *unitary-carrier-square-col*:
assumes *uA*: *unitary A*
shows *A* ∈ *carrier-mat* (*dim-col A*) (*dim-col A*)
⟨*proof*⟩

lemma *unitary-carrier-square-row*:
assumes *uA*: *unitary A*
shows *A* ∈ *carrier-mat* (*dim-row A*) (*dim-row A*)
⟨*proof*⟩

lemma *unitary-dagger*:
assumes *uA*: *unitary A*
shows *unitary* $(A^\dagger)$
⟨*proof*⟩

lemma *unitary-mult*:
assumes *UA*: *unitary A*
and *UB*: *unitary B*
and *dimA*: $A \in$ *carrier-mat n n*
and *dimB*: $B \in$ *carrier-mat n n*
shows *unitary* $(A * B)$
⟨*proof*⟩

9 Kronecker products

Kronecker products of matrices, $A \bigotimes B$: the tensor of identities, the adjoint $(R \otimes T)^* = R^* \otimes T^*$, the mixed-product property $(R_1 \otimes T_1)(R_2 \otimes T_2) = R_1 R_2 \otimes T_1 T_2$, and unitarity of tensors of unitaries.

9.1 Tensor of Identities

lemma *tensor-mat-id*:
Tensor.tensor-mat $(1_m\ n)\ (1_m\ m) = 1_m\ (n * m)$
⟨*proof*⟩

9.2 Mixed-Product Property

lemma *tensor-mat-mult*:
assumes *dimAC*: *dim-col A* = *dim-row C*
and *dimBD*: *dim-col B* = *dim-row D*
shows *Tensor.tensor-mat A B* ∗ *Tensor.tensor-mat C D* =
Tensor.tensor-mat $(A * C)\ (B * D)$
⟨*proof*⟩

9.3 Corollaries of the Mixed-Product Property

lemma *tensor-adjoint-mult*:
(*Tensor.tensor-mat* $(A^\dagger)\ (B^\dagger)$) ∗ (*Tensor.tensor-mat A B*) =
Tensor.tensor-mat $((A^\dagger) * A)\ ((B^\dagger) * B)$
⟨*proof*⟩

lemma *tensor-mult-adjoint*:
(*Tensor.tensor-mat A B*) ∗ (*Tensor.tensor-mat* $(A^\dagger)\ (B^\dagger)$) =
Tensor.tensor-mat $(A * (A^\dagger))\ (B * (B^\dagger))$
⟨*proof*⟩

9.4 Tensor Unitarity

lemma *tensor-mat-unitary-carrier*:
assumes *dimA*: $A \in$ *carrier-mat n n*
and *dimB*: $B \in$ *carrier-mat m m*
and *uA*: *unitary A*
and *uB*: *unitary B*
shows *unitary* (*Tensor.tensor-mat A B*)
⟨*proof*⟩

lemma *tensor-mat-unitary*:

assumes *uA*: *unitary A*
and *uB*: *unitary B*
shows *unitary* (*Tensor.tensor-mat A B*)
⟨*proof*⟩

lemma *tensor-mat-unitary-one-left* [*simp*]:
assumes *uA*: *unitary A*
shows *unitary* (*Tensor.tensor-mat* (1_m *n*) *A*)
⟨*proof*⟩

lemma *tensor-mat-unitary-one-right* [*simp*]:
assumes *uA*: *unitary A*
shows *unitary* (*Tensor.tensor-mat A* (1_m *n*))
⟨*proof*⟩

lemma *one-smult-mat*: (*1*::*complex*) $\cdot_m$ *M* = *M*
⟨*proof*⟩

lemma *mult-index-lt*:
fixes *i j n m* :: *nat*
assumes *i* < *n* **and** *j* < *m*
shows *i* ∗ *m* + *j* < *n* ∗ *m*
⟨*proof*⟩

lemma *mult-index-div-mod*:
fixes *i l m* :: *nat*
assumes *l* < *m*
shows (*i* ∗ *m* + *l*) *div m* = *i* (*i* ∗ *m* + *l*) *mod m* = *l*
⟨*proof*⟩

lemma *mult-index-eq-iff*:
fixes *i j k l m* :: *nat*
assumes *k* < *m* **and** *l* < *m*
shows *i* ∗ *m* + *k* = *j* ∗ *m* + *l* ⟷ *i* = *j* ∧ *k* = *l*
⟨*proof*⟩

9.5 Tensor products of matrices

lemma *tensor-index-c*:
fixes *A B* :: *complex Matrix.mat*
assumes *A* ∈ *carrier-mat r c B* ∈ *carrier-mat p q i* < *r* ∗ *p j* < *c* ∗ *q* 0 < *c* 0 < *q*
shows *Tensor.tensor-mat A B* $$ (*i*,*j*) = *A* $$ (*i div p*, *j div q*) ∗ *B* $$ (*i mod p*, *j mod q*)
⟨*proof*⟩

lemma *tensor-one-left*:
fixes *A* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat m m* **and** *m*: 0 < *m*
shows *Tensor.tensor-mat* (1_m *1*) *A* = *A*
⟨*proof*⟩

lemma *tensor-one-right*:
fixes *A* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat m m* **and** *m*: 0 < *m*
shows *Tensor.tensor-mat A* (1_m *1*) = *A*
⟨*proof*⟩

lemma *tensor-mat-add-right*:
fixes *A B C* :: *complex Matrix.mat*

assumes *Ac*: $A \in$ *carrier-mat r c* **and** *Bc*: $B \in$ *carrier-mat p q* **and** *Cc*: $C \in$ *carrier-mat p q*
and *c*: $0 < c$ **and** *q*: $0 < q$
shows *Tensor.tensor-mat A* $(B + C) =$ *Tensor.tensor-mat A B + Tensor.tensor-mat A C*
⟨*proof*⟩

lemma *tensor-mat-add-left*:
fixes *A B C* :: *complex Matrix.mat*
assumes *Ac*: $A \in$ *carrier-mat r c* **and** *Bc*: $B \in$ *carrier-mat r c* **and** *Cc*: $C \in$ *carrier-mat p q*
and *c*: $0 < c$ **and** *q*: $0 < q$
shows *Tensor.tensor-mat* $(A + B)$ $C =$ *Tensor.tensor-mat A C + Tensor.tensor-mat B C*
⟨*proof*⟩

lemma *tensor-mat-smult-right*:
fixes *A B* :: *complex Matrix.mat*
assumes *Ac*: $A \in$ *carrier-mat r c* **and** *Bc*: $B \in$ *carrier-mat p q* **and** *c*: $0 < c$ **and** *q*: $0 < q$
shows *Tensor.tensor-mat A* $(k \cdot_m B) = k \cdot_m$ *Tensor.tensor-mat A B*
⟨*proof*⟩

lemma *tensor-mat-smult-left*:
fixes *A B* :: *complex Matrix.mat*
assumes *Ac*: $A \in$ *carrier-mat r c* **and** *Bc*: $B \in$ *carrier-mat p q* **and** *c*: $0 < c$ **and** *q*: $0 < q$
shows *Tensor.tensor-mat* $(k \cdot_m A)$ $B = k \cdot_m$ *Tensor.tensor-mat A B*
⟨*proof*⟩

10 Typed-dimension matrix wrapper

The type ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* represents matrices of size $CARD('m) \times CARD('n)$ with entries in '*a*, by typedef of the JNF matrix type at the appropriate dimension.

typedef (**overloaded**) ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* =
$\{M$ :: '*a Matrix.mat*. *dim-row* $M = CARD('m) \wedge$ *dim-col* $M = CARD('n)\}$
morphisms *unwrap-lm wrap-lm*
⟨*proof*⟩

setup-lifting *type-definition-lmat*

lemma *unwrap-lm-dim-row* [*simp*]:
dim-row (*unwrap-lm* (M :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)) $= CARD('m)$
⟨*proof*⟩

lemma *unwrap-lm-dim-col* [*simp*]:
dim-col (*unwrap-lm* (M :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)) $= CARD('n)$
⟨*proof*⟩

lemma *unwrap-lm-carrier* [*simp*]:
unwrap-lm (M :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)
$\in$ *carrier-mat* $(CARD('m))$ $(CARD('n))$
⟨*proof*⟩

11 Typeclass instances

The unique zero matrix of the right shape.

instantiation *lmat* :: (*zero*, *finite*, *finite*) *zero*
begin
lift-definition *zero-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*

is 0_m $(CARD('m))$ $(CARD('n))$ ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-zero* [*simp*]:
unwrap-lm (*0* :: ('*a* :: *zero*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)
= 0_m $(CARD('m))$ $(CARD('n))$
⟨*proof*⟩

instantiation *lmat* :: (*plus*, *finite*, *finite*) *plus*
begin
lift-definition *plus-lmat* ::
('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
is (+) ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-plus* [*simp*]:
unwrap-lm ((*A* :: ('*a* :: *plus*, '*m* :: *finite*, '*n* :: *finite*) *lmat*) + *B*)
= *unwrap-lm A* + *unwrap-lm B*
⟨*proof*⟩

instantiation *lmat* :: (*uminus*, *finite*, *finite*) *uminus*
begin
lift-definition *uminus-lmat* ::
('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
is *uminus* ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-uminus* [*simp*]:
unwrap-lm (− (*A* :: ('*a* :: *uminus*, '*m* :: *finite*, '*n* :: *finite*) *lmat*))
= − *unwrap-lm A*
⟨*proof*⟩

instantiation *lmat* :: (*minus*, *finite*, *finite*) *minus*
begin
lift-definition *minus-lmat* ::
('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
is (−) ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-minus* [*simp*]:
unwrap-lm ((*A* :: ('*a* :: *minus*, '*m* :: *finite*, '*n* :: *finite*) *lmat*) − *B*)
= *unwrap-lm A* − *unwrap-lm B*
⟨*proof*⟩

Lift the additive structure laws via $[\![\bigwedge i\ j.\ [\![i < \textit{dim-row ?B};\ j < \textit{dim-col ?B}]\!] \Longrightarrow \textit{?A \$\$ (i, j)} = \textit{?B \$\$ (i, j)};\ \textit{dim-row ?A} = \textit{dim-row ?B};\ \textit{dim-col ?A} = \textit{dim-col ?B}]\!] \Longrightarrow \textit{?A} = \textit{?B}$ on the underlying JNF matrices.

instance *lmat* :: (*semigroup-add*, *finite*, *finite*) *semigroup-add*
⟨*proof*⟩

instance *lmat* :: (*ab-semigroup-add*, *finite*, *finite*) *ab-semigroup-add*
⟨*proof*⟩

instance *lmat* :: (*monoid-add*, *finite*, *finite*) *monoid-add*
⟨*proof*⟩

instance *lmat* :: (*comm-monoid-add*, *finite*, *finite*) *comm-monoid-add*
⟨*proof*⟩

instance *lmat* :: (*group-add*, *finite*, *finite*) *group-add*
⟨*proof*⟩

instance *lmat* :: (*ab-group-add*, *finite*, *finite*) *ab-group-add*
⟨*proof*⟩

12 Real scalar multiplication and Hahn-Banach overloading

HOL's $(\cdot)$ is the abstract real-scalar multiplication used by the Hahn-Banach locales. We overload it on the type (*complex*, *'m*, *'n*) *lmat* as multiplication by the embedded real, lifted from JNF's $(\cdot_m)$.

overloading
prod-lmat ≡
Vector-Space.prod ::
real ⇒ (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
⇒ (*complex*, *'m*, *'n*) *lmat*
begin
definition *prod-lmat* ::
real ⇒ (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
⇒ (*complex*, *'m*, *'n*) *lmat*
where
prod-lmat r M = *wrap-lm* (*complex-of-real r* $\cdot_m$ *unwrap-lm M*)
end

lemma *unwrap-lm-prod* [*simp*]:
unwrap-lm (*Vector-Space.prod r* (*M* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*))
= *complex-of-real r* $\cdot_m$ *unwrap-lm M*
⟨*proof*⟩

13 Real-vectorspace locale on the full type

Every (*complex*, *'m*, *'n*) *lmat* with the universe set is a real vector space in HOL/Hahn_Banach's sense.

lemma *vectorspace-UNIV-lmat*:
Vector-Space.vectorspace (*UNIV* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*)
⟨*proof*⟩

14 The Hilbert-Schmidt norm on (*complex*, *'m*, *'n*) *lmat*

The Hilbert–Schmidt (Frobenius) norm $\|M\| = (\sum_{i,j} |M_{ij}|^2)^{1/2}$, the Euclidean norm of the entries.

definition *hs-norm-lmat* ::
(*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ *real*
where
hs-norm-lmat M =
L2-set (λ(*i*, *j*). *cmod* (*unwrap-lm M* $$ (*i*, *j*))) ({*0*..<*CARD*(*'m*)} × {*0*..<*CARD*(*'n*)})

lemma *hs-norm-lmat-nonneg*: $0 \leq$ *hs-norm-lmat M*
⟨*proof*⟩

Scaling every entry in modulus by $r \geq 0$ scales the norm by r.

lemma *hs-norm-lmat-scale*:
fixes *M N* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
assumes $\bigwedge i\ j.\ i < CARD('m) \Longrightarrow j < CARD('n) \Longrightarrow$
cmod (*unwrap-lm N $$ (i, j)*) = *r* ∗ *cmod* (*unwrap-lm M $$ (i, j)*)
and $0 \leq r$
shows *hs-norm-lmat N* = *r* ∗ *hs-norm-lmat M*
⟨*proof*⟩

lemma *hs-norm-lmat-smult*:
hs-norm-lmat (*Vector-Space.prod a M*) = |*a*| ∗ *hs-norm-lmat M*
⟨*proof*⟩

lemma *hs-norm-lmat-triangle*:
fixes *M N* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
shows *hs-norm-lmat* (*M* + *N*) ≤ *hs-norm-lmat M* + *hs-norm-lmat N*
⟨*proof*⟩

15 Subspace, seminorm, normed-vectorspace instances

HS norm is a seminorm on the whole real vector space of (*complex*, *'m*, *'n*) *lmat*.

lemma *seminorm-UNIV-hs-norm-lmat*:
seminorm (*UNIV* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*) *hs-norm-lmat*
⟨*proof*⟩

HS norm is a (real) norm: it vanishes only on the zero matrix.

lemma *hs-norm-lmat-zero-iff*:
fixes *M* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
shows (*hs-norm-lmat M* = *0*) ⟷ (*M* = *0*)
⟨*proof*⟩

lemma *norm-UNIV-hs-norm-lmat*:
Normed-Space.norm
(*UNIV* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*) *hs-norm-lmat*
⟨*proof*⟩

lemma *normed-vectorspace-UNIV-lmat*:
normed-vectorspace
(*UNIV* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*) *hs-norm-lmat*
⟨*proof*⟩

A complex-linear subspace V of (*complex*, *'m*, *'n*) *lmat* (i.e. nonempty, closed under addition and complex-scalar multiplication) is in particular a real-linear subspace, hence a *subspace* of *UNIV* in HOL/Hahn_Banach's sense.

definition *cmat-subspace-lmat* ::
(*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat set* ⇒ *bool*
where
cmat-subspace-lmat V ≡
V ≠ {} ∧
(∀ *x* ∈ *V*. ∀ *y* ∈ *V*. *x* + *y* ∈ *V*) ∧
(∀ *x* ∈ *V*. ∀ *c* :: *complex*.
wrap-lm (*c* ·$_m$ *unwrap-lm x*) ∈ *V*)

lemma *subspace-of-cmat-subspace-lmat*:
fixes V :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat set*
assumes *cmat-subspace-lmat V*
shows *Subspace.subspace V* (*UNIV* :: (*complex*, $'m$, $'n$) *lmat set*)
⟨*proof*⟩

16 Complex scalar multiplication on lmat

Complex scalar multiplication on (*complex*, $'m$, $'n$) *lmat*, extending the *Vector-Space.prod* overload (which is only by reals).

definition *cmult-lmat* ::
complex ⇒ (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
⇒ (*complex*, $'m$, $'n$) *lmat*
where
cmult-lmat c M = *wrap-lm* (*c* ·$_m$ *unwrap-lm M*)

lemma *unwrap-lm-cmult* [*simp*]:
unwrap-lm (*cmult-lmat c* (*M* :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*))
= *c* ·$_m$ *unwrap-lm M*
⟨*proof*⟩

lemma *cmult-lmat-real*:
cmult-lmat (*complex-of-real a*) *M* = *Vector-Space.prod a M*
⟨*proof*⟩

lemma *cmult-lmat-add*:
fixes *x y* :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
shows *cmult-lmat c* (*x* + *y*) = *cmult-lmat c x* + *cmult-lmat c y*
⟨*proof*⟩

lemma *cmult-lmat-scalar-add*:
fixes *x* :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
shows *cmult-lmat* (*c* + *d*) *x* = *cmult-lmat c x* + *cmult-lmat d x*
⟨*proof*⟩

lemma *cmult-lmat-scalar-mult*:
fixes *x* :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
shows *cmult-lmat* (*c* ∗ *d*) *x* = *cmult-lmat c* (*cmult-lmat d x*)
⟨*proof*⟩

lemma *cmult-lmat-uminus-one*:
fixes *x* :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
shows *cmult-lmat* (− *1*) *x* = − *x*
⟨*proof*⟩

lemma *cmult-lmat-i-sq*:
fixes *x* :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
shows *cmult-lmat* i (*cmult-lmat* i *x*) = − *x*
⟨*proof*⟩

lemma *hs-norm-lmat-cmult*:
fixes *M* :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
shows *hs-norm-lmat* (*cmult-lmat c M*) = *cmod c* ∗ *hs-norm-lmat M*
⟨*proof*⟩

The HOL/Hahn_Banach development imports locales also called *norm*, *seminorm* and *continuous*. Hide their short names so that *norm* is the class method from *HOL.Real-Vector-Spaces* and *continuous* the topological predicate.

hide-const (**open**) *Normed-Space.norm*
hide-const (**open**) *Normed-Space.seminorm*
hide-const (**open**) *Function-Norm.continuous*

17 Overview

Nothing in this theory is stated in the notes; it is the bridge that lets the HOL-Analysis theorems on convex sets (separating hyperplanes, compactness) be applied to matrices. (*'a*, *'m*, *'n*) *lmat* is given the typeclass tower *real-vector*, *metric-space*, *real-normed-vector*, *real-inner*, *euclidean-space*, *heine-borel*; for *'a* = *complex* the norm is the Hilbert–Schmidt norm, the inner product its real part, and the dimension is $2\,\mathrm{CARD}('m)\,\mathrm{CARD}('n)$.

18 Real scalar multiplication and *real-vector* instance

For *'a* :: *real-vector*, lift element-wise scalar multiplication via *map-mat* (*scaleR a*).

instantiation *lmat* :: (*real-vector*, *finite*, *finite*) *scaleR*
begin

lift-definition *scaleR-lmat* ::
  real ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  is λ*r M*. *map-mat* (λ*x*. *r* ∗$_R$ *x*) *M* ⟨*proof*⟩

instance ⟨*proof*⟩

end

lemma *unwrap-lm-scaleR* [*simp*]:
  unwrap-lm ((*r* :: *real*) ∗$_R$ (*M* :: (*'a* :: *real-vector*, *'m* :: *finite*, *'n* :: *finite*) *lmat*))
    = *map-mat* (λ*x*. *r* ∗$_R$ *x*) (*unwrap-lm M*)
  ⟨*proof*⟩

lemma *unwrap-lm-scaleR-index* [*simp*]:
  fixes *M* :: (*'a* :: *real-vector*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  assumes *i* < *CARD*(*'m*) *j* < *CARD*(*'n*)
  shows *unwrap-lm* (*r* ∗$_R$ *M*) $$ (*i*, *j*) = *r* ∗$_R$ (*unwrap-lm M* $$ (*i*, *j*))
  ⟨*proof*⟩

instance *lmat* :: (*real-vector*, *finite*, *finite*) *real-vector*
⟨*proof*⟩

19 Entries of sums and differences

lemma *lmat-eq-iff-entries*:
  fixes *M N* :: (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  shows *M* = *N* ⟷
      (∀ *i j*. *i* < *CARD*(*'m*) ⟶ *j* < *CARD*(*'n*)
          ⟶ *unwrap-lm M* $$ (*i*, *j*) = *unwrap-lm N* $$ (*i*, *j*))
⟨*proof*⟩

20 Distance, uniformity, open: *metric-space* instance

Distance is the L2 sum of entry-wise distances, indexed by $\{0..<CARD('m)\} \times \{0..<CARD('n)\}$. Uniformity and *open* are defined via the standard metric-induced filter so that the *open-uniformity* axiom is discharged by reflexivity.

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *dist*
begin

definition *dist-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ *real* **where**
dist-lmat M N = *L2-set* (λ(*i*, *j*). *dist* (*unwrap-lm M* $$ (*i*, *j*)) (*unwrap-lm N* $$ (*i*, *j*)))
({*0..<CARD('m)*} × {*0..<CARD('n)*})

instance ⟨*proof*⟩

end

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *uniformity-dist*
begin

definition *uniformity-lmat* ::
(('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* × ('*a*, '*m*, '*n*) *lmat*) *filter* **where**
(*uniformity-lmat* :: (('*a*, '*m*, '*n*) *lmat* × ('*a*, '*m*, '*n*) *lmat*) *filter*)
= (*INF e* ∈ {*0<..*}. *principal* {(*x*, *y*). *dist x y* < *e*})

instance ⟨*proof*⟩

end

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *metric-space*
begin

definition *open-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat set* ⇒ *bool* **where**
open-lmat U = (∀ *x* ∈ *U*. ∀ *F* (*x'*, *y*) *in uniformity*. *x'* = *x* ⟶ *y* ∈ *U*)

instance ⟨*proof*⟩

end

21 Norm and *real-normed-vector* instance

instantiation *lmat* :: (*real-normed-vector*, *finite*, *finite*) *real-normed-vector*
begin

definition *norm-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ *real* **where**
norm-lmat M = *L2-set* (λ(*i*, *j*). *norm* (*unwrap-lm M* $$ (*i*, *j*)))
({*0..<CARD('m)*} × {*0..<CARD('n)*})

definition *sgn-lmat* ::
('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ ('*a*, '*m*, '*n*) *lmat* **where**
sgn-lmat (*M* :: ('*a*, '*m*, '*n*) *lmat*) = *scaleR* (*inverse* (*norm M*)) *M*

instance ⟨*proof*⟩

end

22 Real inner product

For $'a :: \mathit{real\text{-}inner}$, the lmat inner product is the entry-wise sum of inner products. When $'a = \mathit{complex}$ with the standard real-inner instance $\mathit{inner}\ z\ w = \mathit{Re}\ (\mathit{cnj}\ z * w)$, this is the real part of the Hilbert-Schmidt inner product.

instantiation *lmat* :: (*real-inner*, *finite*, *finite*) *real-inner*
begin

definition *inner-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
⇒ ('*a*, '*m*, '*n*) *lmat* ⇒ *real* **where**
inner-lmat M N = (∑ (*i*, *j*) ∈ {*0*..<*CARD*('*m*)} × {*0*..<*CARD*('*n*)}.
inner (*unwrap-lm M* $$ (*i*, *j*)) (*unwrap-lm N* $$ (*i*, *j*)))

instance ⟨*proof*⟩

end

23 Basis: matrix units carrying a basis vector

definition *lmat-axis* :: *nat* ⇒ *nat* ⇒ '*a* :: *zero*
⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* **where**
lmat-axis i j u = *wrap-lm* (*Matrix.mat* (*CARD*('*m*)) (*CARD*('*n*))
(λ(*r*, *s*). *if r* = *i* ∧ *s* = *j then u else 0*))

lemma *unwrap-lmat-axis* [*simp*]:
unwrap-lm (*lmat-axis i j u* :: ('*a* :: *zero*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)
= *Matrix.mat* (*CARD*('*m*)) (*CARD*('*n*))
(λ(*r*, *s*). *if r* = *i* ∧ *s* = *j then u else 0*)
⟨*proof*⟩

Inner product against a single matrix-unit basis vector picks out one matrix entry.

lemma *inner-lmat-axis*:
fixes *u* :: '*a* :: *real-inner*
and *M* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes *ij*: *i* < *CARD*('*m*) *j* < *CARD*('*n*)
shows *inner M* (*lmat-axis i j u* :: ('*a*, '*m*, '*n*) *lmat*)
= *inner* (*unwrap-lm M* $$ (*i*, *j*)) *u*
⟨*proof*⟩

Two matrix-unit basis vectors are equal iff their entries and positions match (or both vectors are zero).

lemma *lmat-axis-eq-iff*:
fixes *u v* :: '*a* :: *zero*
assumes *i* < *CARD*('*m* :: *finite*) *j* < *CARD*('*n* :: *finite*)
and *i'* < *CARD*('*m*) *j'* < *CARD*('*n*)
shows (*lmat-axis i j u* :: ('*a*, '*m*, '*n*) *lmat*) = *lmat-axis i' j' v*
⟷ (*u* = *v* ∧ *i* = *i'* ∧ *j* = *j'*) ∨ (*u* = *0* ∧ *v* = *0*)
⟨*proof*⟩

Inner product between two matrix-unit basis vectors: 1 if same position and basis component, 0 otherwise.

lemma *inner-lmat-axis-axis*:
fixes *u v* :: '*a* :: *real-inner*
assumes *ij*: *i* < *CARD*('*m* :: *finite*) *j* < *CARD*('*n* :: *finite*)
and *ij'*: *i'* < *CARD*('*m*) *j'* < *CARD*('*n*)
shows *inner* (*lmat-axis i j u* :: ('*a*, '*m*, '*n*) *lmat*) (*lmat-axis i' j' v*)

$= (\mathit{if}\ i = i' \wedge j = j'\ \mathit{then}\ \mathit{inner}\ u\ v\ \mathit{else}\ 0)$
⟨*proof*⟩

24 Basis set

definition *Basis-lmat-set* :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*
  where *Basis-lmat-set* = ($\bigcup i \in \{0..<CARD('m)\}$.
    $\bigcup j \in \{0..<CARD('n)\}$.
      $\bigcup u \in Basis$. {*lmat-axis i j u* :: (*'a*, *'m*, *'n*) *lmat*})

lemma *Basis-lmat-set-finite*:
  finite (*Basis-lmat-set* :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*)
  ⟨*proof*⟩

lemma *Basis-lmat-set-nonempty*:
  Basis-lmat-set ≠ ({} :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*)
⟨*proof*⟩

lemma *Basis-lmat-orthonormal*:
  fixes *u v* :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  assumes *uB*: *u* ∈ *Basis-lmat-set* **and** *vB*: *v* ∈ *Basis-lmat-set*
  shows *inner u v* = (*if u* = *v then 1 else 0*)
⟨*proof*⟩

lemma *axis-mem-Basis-lmat-set*:
  fixes *u* :: *'a* :: *euclidean-space*
  assumes *i-lt*: *i* < *CARD*(*'m* :: *finite*)
    and *j-lt*: *j* < *CARD*(*'n* :: *finite*)
    and *uB*: *u* ∈ *Basis*
  shows (*lmat-axis i j u* :: (*'a*, *'m*, *'n*) *lmat*) ∈ *Basis-lmat-set*
⟨*proof*⟩

lemma *Basis-lmat-perp-imp-zero*:
  fixes *x* :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  assumes ∗: ∀ *u* ∈ *Basis-lmat-set*. *inner x u* = *0*
  shows *x* = *0*
⟨*proof*⟩

lemma *zero-inner-Basis-lmat*:
  fixes *u* :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  shows *inner* (*0* :: (*'a*, *'m*, *'n*) *lmat*) *u* = *0*
  ⟨*proof*⟩

lemma *Basis-lmat-perp-zero*:
  fixes *x* :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  shows (∀ *u* ∈ *Basis-lmat-set*. *inner x u* = *0*) ⟷ *x* = *0*
  ⟨*proof*⟩

25 Euclidean space instance

instantiation *lmat* :: (*euclidean-space*, *finite*, *finite*) *euclidean-space*
begin

definition *Basis-lmat* :: (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat set* **where**
  Basis-lmat = *Basis-lmat-set*

instance ⟨*proof*⟩

end

26 Heine-Borel

instance *lmat* :: (*euclidean-space*, *finite*, *finite*) *heine-borel* ⟨*proof*⟩

27 Dimension

Two distinct (i, j, u) with $u \in$ *Basis-'a* give distinct matrix-axis vectors.

lemma *lmat-axis-inj-on-Basis-indices*:
  fixes *u v* :: *'a* :: *euclidean-space*
  assumes *ij*: $i < CARD('m :: finite)$ $j < CARD('n :: finite)$
    and *ij'*: $i' < CARD('m)$ $j' < CARD('n)$
    and *uB*: $u \in Basis$ **and** *vB*: $v \in Basis$
  shows (*lmat-axis i j u* :: (*'a*, *'m*, *'n*) *lmat*) = *lmat-axis i' j' v*
    $\longleftrightarrow i = i' \wedge j = j' \wedge u = v$
⟨*proof*⟩

lemma *DIM-lmat*:
  $DIM(('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat)$
    $= CARD('m) * CARD('n) * DIM('a)$
⟨*proof*⟩

28 Component-wise bounds via the L2 norm

lemma *norm-lmat-index-le*:
  fixes *M* :: (*'a* :: *real-normed-vector*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  assumes $i < CARD('m)$ $j < CARD('n)$
  shows *norm* (*unwrap-lm M $$ (i, j)*) ≤ *norm M*
⟨*proof*⟩

lemma *dist-lmat-index-le*:
  fixes *M N* :: (*'a* :: *real-normed-vector*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  assumes $i < CARD('m)$ $j < CARD('n)$
  shows *dist* (*unwrap-lm M $$ (i, j)*) (*unwrap-lm N $$ (i, j)*) ≤ *dist M N*
⟨*proof*⟩

29 Convergence: lmat sequence convergence implies entry convergence

lemma *tendsto-unwrap-lm-index*:
  fixes *f* :: *'b* ⇒ (*'a* :: *real-normed-vector*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
    and *L* :: (*'a*, *'m*, *'n*) *lmat*
  assumes *lim*: (*f* ⟶ *L*) *F*
    and *ij*: $i < CARD('m)$ $j < CARD('n)$
  shows ((λ*x*. *unwrap-lm* (*f x*) $$ (*i*, *j*)) ⟶ *unwrap-lm L $$ (i, j)*) *F*
⟨*proof*⟩

30 Entries as continuous functions; wrapping

The entry-projection $M \mapsto$ *unwrap-lm M $$ (i, j)* is continuous on lmat for valid index (i, j).

lemma *isCont-unwrap-lm-index*:

fixes x :: ($'a$ :: *real-normed-vector*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
assumes *ij*: $i < CARD('m)$ $j < CARD('n)$
shows *isCont* (λM. *unwrap-lm* M $$ (i, j)) x
⟨*proof*⟩

lemma *continuous-on-unwrap-lm-index*:
fixes S :: ($'a$ :: *real-normed-vector*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat set*
assumes *ij*: $i < CARD('m)$ $j < CARD('n)$
shows *continuous-on* S (λM. *unwrap-lm* M $$ (i, j))
⟨*proof*⟩

lemma *continuous-on-unwrap-lm-index-comp* [*continuous-intros*]:
fixes f :: $'b$ :: *topological-space* ⇒ ($'a$ :: *real-normed-vector*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
assumes *continuous-on* S f **and** $i < CARD('m)$ **and** $j < CARD('n)$
shows *continuous-on* S (λx. *unwrap-lm* (f x) $$ (i, j))
⟨*proof*⟩

Conversely, a map into lmat is continuous when all its entries are.

lemma *continuous-on-lmat-entries*:
fixes f :: $'b$ :: *t2-space* ⇒ ($'a$ :: *euclidean-space*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
assumes $\bigwedge i\ j.\ i < CARD('m) \Longrightarrow j < CARD('n) \Longrightarrow$ *continuous-on* S (λx. *unwrap-lm* (f x) $$ (i, j))
shows *continuous-on* S f
⟨*proof*⟩

lemma *unwrap-wrap-lm*:
fixes M :: *complex Matrix.mat*
assumes $M \in$ *carrier-mat CARD*($'k$::*finite*) *CARD*($'k$)
shows *unwrap-lm* (*wrap-lm* M :: (*complex*, $'k$, $'k$) *lmat*) = M
⟨*proof*⟩

lemma *unwrap-lm-scaleR-mat*:
fixes R :: (*complex*, $'k$ :: *finite*, $'k$) *lmat*
shows *unwrap-lm* (u $*_R$ R) = *complex-of-real* u $\cdot_m$ *unwrap-lm* R
⟨*proof*⟩

31 The bundle at a glance

Four theories, none of which depends on the lectures.

31.1 Matrix lemmas

Sums, adjoints and index arithmetic for *complex Matrix.mat*: finite sums of matrices (*msum*, *foldr*-sums), entries of products, linear maps on a set of matrices and linear-space isomorphisms.

⟦*?Y* ∈ *carrier-mat ?p ?q*; *?i* < *?q*; *?j* < *?q*⟧
⟹ (*?Y*† ∗ *?Y*) $$ (*?i*, *?j*) = ⟨*Matrix.col ?Y ?i*|*Matrix.col ?Y ?j*⟩

⟦*?M* ∈ *carrier-mat ?k ?k*; *?N* ∈ *carrier-mat ?k ?k*;
⋀*x*. *x* ∈ *carrier-vec ?k* ⟹ *?M* $*_v$ *x* = *?N* $*_v$ *x*⟧
⟹ *?M* = *?N*

⟦*?A* ∈ *carrier-mat ?m ?r*; ∀ *x*∈*set ?xs*. *?g x* ∈ *carrier-mat ?r ?c*⟧
⟹ *?A* ∗ *msum ?r ?c ?g ?xs* = *msum ?m ?c* (λ*x*. *?A* ∗ *?g x*) *?xs*

$[\![\forall x{\in}set\ ?xs.\ ?g\ x \in carrier\text{-}mat\ ?r\ ?c;\ ?B \in carrier\text{-}mat\ ?c\ ?p]\!]$
$\Longrightarrow msum\ ?r\ ?c\ ?g\ ?xs * ?B = msum\ ?r\ ?p\ (\lambda x.\ ?g\ x * ?B)\ ?xs$

$(?x = ?y) = (?x\ div\ ?k = ?y\ div\ ?k \wedge ?x\ mod\ ?k = ?y\ mod\ ?k)$

$[\![lin\text{-}on\ ?n\ ?d\ ?S\ ?\Phi;\ 0_m\ ?n\ ?n \in ?S;\ \bigwedge A\ B.\ [\![A \in ?S;\ B \in ?S]\!] \Longrightarrow A + B \in ?S;$
$\forall x{\in}set\ ?xs.\ ?g\ x \in ?S]\!]$
$\Longrightarrow ?\Phi\ (msum\ ?n\ ?n\ ?g\ ?xs) = msum\ ?d\ ?d\ (\lambda x.\ ?\Phi\ (?g\ x))\ ?xs$

31.2 Unitary and Kronecker-product algebra

Unitary matrices are closed under products and adjoints; the Kronecker product satisfies the mixed-product law, so tensor products of unitaries are unitary.

$unitary\ ?A \Longrightarrow unitary\ ?A^{\dagger}$

$[\![unitary\ ?A;\ unitary\ ?B;\ ?A \in carrier\text{-}mat\ ?n\ ?n;\ ?B \in carrier\text{-}mat\ ?n\ ?n]\!]$
$\Longrightarrow unitary\ (?A * ?B)$

$[\![dim\text{-}col\ ?A = dim\text{-}row\ ?C;\ dim\text{-}col\ ?B = dim\text{-}row\ ?D]\!]$
$\Longrightarrow (?A \bigotimes ?B) * (?C \bigotimes ?D) = ?A * ?C \bigotimes ?B * ?D$

$[\![unitary\ ?A;\ unitary\ ?B]\!] \Longrightarrow unitary\ (?A \bigotimes ?B)$

$?l < ?m \Longrightarrow (?i * ?m + ?l)\ div\ ?m = ?i$
$?l < ?m \Longrightarrow (?i * ?m + ?l)\ mod\ ?m = ?l$

31.3 Matrices of fixed dimension

$('a,\ 'm,\ 'n)\ lmat$ carries the dimensions in its type. With the Hilbert–Schmidt norm it is a normed vector space in the sense of *HOL−Hahn-Banach*.

$hs\text{-}norm\text{-}lmat\ (?M + ?N) \leq hs\text{-}norm\text{-}lmat\ ?M + hs\text{-}norm\text{-}lmat\ ?N$

$(hs\text{-}norm\text{-}lmat\ ?M = 0) = (?M = 0)$

$normed\text{-}vectorspace\ UNIV\ hs\text{-}norm\text{-}lmat$

31.4 Matrices as a Euclidean space

For $'a = complex$ the type $lmat$ is a *euclidean-space* of dimension $2\,\mathrm{CARD}('m)\,\mathrm{CARD}('n)$, so the convexity and compactness results of *HOL−Analysis* apply to matrices; entries depend continuously on the matrix.

$DIM((?'a,\ ?'m,\ ?'n)\ lmat) = CARD(?'m) * CARD(?'n) * DIM(?'a)$

$[\![?i < CARD(?'m);\ ?j < CARD(?'n)]\!]$
$\Longrightarrow norm\ (unwrap\text{-}lm\ ?M\ \$\$\ (?i,\ ?j)) \leq norm\ ?M$

$(\bigwedge i\ j.\ [\![i < CARD(?'m);\ j < CARD(?'n)]\!]$
$\quad \Longrightarrow continuous\text{-}on\ ?S\ (\lambda x.\ unwrap\text{-}lm\ (?f\ x)\ \$\$\ (i,\ j))) \Longrightarrow$
$continuous\text{-}on\ ?S\ ?f$